MILITARY SPECIFICATION

BANDOLEER XM 86 MINE

This specification is approved for use by the Natick Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type and size of Bandoleer fabricated from olive nylon duck, webbing and tape, with miscellaneous hardware.

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-268	-	Paper, Kraft, Wrapping
DDD-L-20	-	Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams, and Stitchings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760-5014 by using the Self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1345

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

2-3-448 Bandoleer Assembly XM 86 Mine
2-3-449 Cover pattern
2-3-450 Front pattern
2-3-451 Cover pad
2-3-452 Reinforcement panel

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the test of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. Samples of the end item when furnished, are solely for guidance and information to the contractor see (6.3). Variations from this document may appear in the sample, in which case this document shall govern.

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3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.3 Materials and components. Materials (see 6.6 and 6.7) and components shall be as specified on Drawing add and parts list.

3.4 Construction. The construction shall conform in all respects to the drawings listed in section 2 and as specified herein.

3.4.1 Stitches, seams and stitchings. All stitching, shall conform to FED-STD-751 as follows:

For all stitching	Stitch type 301, 8 to 10 stitches per inch

3.4.1.1 Type 301 stitching. Ends of stitching shall be backstitched or over stitched not less than 1 inch except where ends are turned under or caught in other seams and stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be imbedded in the materials sewn.

3.4.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 1 inch in back of the defective area, continue over the defective area and continue a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.4.1.2 Bartacks. Bartacks shall be as specified on the drawings and as follows:

Length, 1/16	Width, 1/32	Stitches per bartack
1/2 inch	1/8 inch	28
3/4 inch	1/8 inch	42

Bartacks shall be free from thread breaks and loose stitching.

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3.4.1.3 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.4.2 Fusing of ends of nylon tape and webbing. Where specified on drawing, ends of nylon tape and webbing shall be fused. The apparatus used to fuse the tape ends shall be capable of providing sufficient heat to provide a smooth edge and with the cut ends of the tape yarns all fused together.

3.4.3 Setting of snap fasteners. A hole shall be prepunched to receive the button and eyelet components of the snap fasteners. The hole shall be smaller than the outside diameter of the button and eyelet barrels so that the barrel must be forced through the hole. The hole shall not be punched in the setting operation with the button or eyelet barrel. The fasteners shall be securely clinched without cutting the adjacent materials and no more than three splits shall occur in the button or eyelet barrels.

3.4.4 Setting of eyelets. Holes shall be prepunched to receive the eyelets. The holes shall be smaller than the outside diameter of the eyelet barrel so that the barrel must be forced through the hole. The eyelet shall be installed using score setting dies conforming to J. C. Rhodes, Part Number 1-K-1KL and 101Y-K1om or Stimpson Co. Part Number A2744-7 or equal (see 6.5). The eyelet shall be securely set in a manner that will prevent detachment from, or cutting of, the adjacent material. The clinched portion of the eyelet shall be on the inside of the case.

3.5 Marking. The identification marking shall be applied in the location shown on the applicable drawing and shall conform to type IV, class 5 of DDD-L-20. The letters "US" shall be applied in the size characters and in the location indicated on the applicable drawing and shall conform to type IV, class 9 of DDD-L-20. Fastness of class 9 marking shall be as specified for class 5 marking.

3.5.1 Location marks. Location marks shall not be drilled except for locating snap fasteners and eyelets. Printed markings for component location shall not be more than 1/32 inch wide.

3.6 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the bandoleer.

3.7 Replacement of defective components. During the spreading, cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.3, shall be removed from production and replaced with nondefective and properly matched components.

3.8 Workmanship. The finished and assembled bandoleer shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3, 4.4.4 and 4.4.5. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether cut lengths, cut parts, markings for location of components, and location of assembled component parts are in accordance with specified requirements. In addition, inspection shall be made to assure that the prepunched holes for receiving snap fasteners and eyelets are as specified in 3.4.3 and 3.4.4. Whenever nonconformance is noted, corrections shall be made to the parts affected and lot in process. Parts which cannot be corrected shall be removed from production.

4.4.3 End item visual examination. The end item shall be examined for the defects listed in table I. The lot size shall be expressed in units of bandoleers. The sample unit shall be one completely fabricated bandoleer. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 10.0 for total major and minor combined defects.

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TABLE I. End item visual defects

Examine	Defect	Classification Major	Minor
Fabric	Hole, cut, tear, smash, broken or missing yarn, or open place clearly visible at normal inspection distance (approximately 3 feet)	X	
Tape and webbing	Not firmly and tightly woven; edges frayed or scalloped	X	
	Multiple floats		X
	Any cut, hole, tear or smash	X	
	Abrasion mark, slub, broken end or pick		X
	Ends not fused as specified (where required)		X
Metal hardware (general)	Broken or malformed failing to serve intended purpose, corroded area, burr or sharp edge	X	
	Finish omitted, partially omitted or not as specified -on brass or aluminum components		X
	-on steel components	X	
	Not assembled as specified (unless otherwise classified herein)	X	
	Not specified type, size or style	X	
Snap fastener	Fastener not functioning properly i.e., fails to snap closed, provide a secure closure or to open freely	X	
NOTE:	The fastener shall be snapped and unsnapped twice to determine whether parts of fastener separate freely and also effect a secure closure.		
	Clinched excessively tight, cutting adjacent material	X	

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Snap fastener (cont'd)	Clinched loosely, permitting any component to rotate freely but not to the degree that any component can be expected to become detached during use		X
	Clinched loosely to the degree that components can be expected to become detached during use	X	
	Incorrect style	X	
	More than three splits in eyelet or button barrels		X
	NOTE: Incomplete roll of end of button or eyelet barrel is evidence of improper and insecure clinching.		
Eyelet	Insecurely clinched, or clinched excessively tight, cutting fabric	X	
	Installed with scored side on outside of case		X
	Not set with a scored setting		X
Open seam	1/2 inch or less		X
	More than 1/2 inch	X	
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken, or when two or more consecutive skipped or runoff stitches occur.		
Raw edge (on edge required to be finished)	More than 1/2 inch in length when securely caught in stitching	X	
	NOTE: Raw edge not securely caught in stitching shall be classified as an open seam.		

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
	Runoff (see open seam)		
Seam and stitch	Wrong seam or stitch type	X	
Bartacks	One or more bartacks omitted	X	
	Any bartack not as specified or not in specified location		X
	Loose stitching, incomplete or broken		X
Stitch tension	Loose, resulting in a loose bobbin or top thread		X
	Excessively tight, resulting in puckering of material		X
Stitches per inch	One stitch less than minimum specified		X
	Two or more stitches less than minimum specified	X	
	One or more stitches in excess of maximum specified		X
	NOTE: Variation in the number of stitches per inch caused by the operator speeding up the machine and pulling the materials in order to sew over heavy places or heavy seams, or in turning corners shall be classified as follows: (a) Within the minor defect classification - no defect (b) Within the major defect classification - minor defect		
Thread breaks, skipped stitches, or runoffs	Not overstitched as specified		X
	NOTE: Thread breaks or two or more consecutive skipped or runoff stitches not overstitched shall be classified as open seams.		

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TABLE I. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Ends of stitching	Not secured as specified		X
Rows of stitching	Any row missing	X	
Components and assembly	Any component part omitted or not as specified or any required operation omitted (unless otherwise classified herein)	X	
	Needle chews	X	
	Any mend, darn, patch, or splice	X	
Binding	Loosely applied but not exposing raw edge of material		X
	Loosely applied exposing raw edge of material	X	
Marking	Omitted, illegible, incorrect, or misplaced		X
Location markings	Drilled	X	
	Printed marking more than 1/32 inch in width		X
Cleanness	Any spot or stain clearly noticeable		X
Thread ends	Not trimmed to 1/4 inch maximum length		X

4.4.4 Fit examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of bandoleers. The sample unit shall be one bandoleer. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 0.65.

Examine	Defect
Fit of mine into bandoleer and closure of flap	Bandoleers too small, i.e., mine fails to fit properly within the bandoleer.
	The mine shall be inserted into the bandoleer with the pin side up and facing outboard whether in either pocket.

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NOTE: It shall be possible to insert the mine within the bandoleer pocket without effort other than that necessary to overcome friction between the mine and the pocket. A defect shall be scored if the mine must be forced into the pocket.

Bandoleer, pocket body or flap too short, causing inability to secure snap fastener without applying excessive force.

NOTE: With the two mines in the bandoleer pockets, the pocket flaps shall be closed and the snap fasteners, shall be secured. It shall be possible to secure the snap fastener with out effort other than that necessary to secure the male and female components. A defect shall be scored if it is necessary to force either flap down in order to secure the fastener.

4.4.5 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified on drawings 2-3-448, 2-3-451, and 2-3-452. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of bandoleers. The sample unit shall be one bandoleer. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 10.0.

4.4.6 Packaging inspection. An examination shall be made to determine that preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged. The lot size shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete closure of container flap, improper taping, loose strapping, inadequate stapling
	Bulged or distorted container
Content	Number of bandoleers per bundle is more or less than specified 1/
	Number of bundles per container is more or less than specified

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1/ For this defect, one bundle shall be examined from each shipping container in the sample.

4.4.7 Palletization examination. An examination shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified
	Interlocking of loads not as specified
	Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING.

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each bandoleer shall have the flap snap fastener closed. Six bandoleers, alternately reversed end for end, shall be neately stacked to form a bundle. The bundle shall be securely tied in the center with cotton tape or twine.

5.1.2 Commercial. Bandoleers shall be preserved in accordance with ASTM D-3951.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2).

5.2.1 Level A packing. One-hundred forty four bandoleers, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. The inside dimensions of the shipping container shall approximate 16 inches in length, 12 inches in width and 18 inches in depth. Approximate dimensions are furnished

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as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that inspection shall be in accordance with 4.4.6.

5.2.2 Level B packing. One-hundred forty four bandoleers, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside dimensions of the shipping container shall approximate 16 inches in length, 12 inches in width and 18 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that inspection shall be in accordance with 4.4.6

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.6.

5.2.3 Commercial packing. Bandoleers, preserved as specified in 5.1, shall be packed in accordance with ASTM D-3951.

5.3 Palletization. When specified (see 6.2), bandoleers, packed as specified in 5.2, shall be palletized in accordance with load type 1a of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D-3951, as applicable.

6. NOTES

6.1 Intended use. The bandoleer is designed to accommodate and carry two XM 86 mines. The bandoleer is provided with a support strap to be worn over the shoulder and a leg tie down loop located on the bottom. The bandoleer is of the expendable type.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, name, and date of this document.

b. When a first article sample is required (see 3.1, 4.3, and 6.4).

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c. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).

d. When weather-resistant grade shipping containers are required for level B packing (see 5.2.2.1).

e. When palletization is required (see 5.3).

6.3 Samples. For access to samples and the mine required for the fit examination in 4.4.4 address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample consisting of one completed bandoleer. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.5 Die, eyelet, score setting. Sources of supply for the score setting eyelet die specified in 3.4.4 are, J. C. Rhodes and Company, Box B-962, New Bedford, MA 02745 and Stimpson Co., Bayport, NY 11705.

6.6 Nylon cloth cut edges. The cut edges of the nylon cloth are subject to fraying during the process of handling the cut parts during fabrication of the bandoleer. It has been found that fusing of the cut edges of the cloth will prevent fraying.

6.7 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

Custodian:

Army - GL

Preparing activity:

Army - GL

Project No. 1345-A292

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44199(GL)	Bandoleer XM 86 Mine

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 20 Nov 84

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